

Illustration by TANMOY CHAKRABORTY

NIPAH AND HENIPAVIRUS DISEASES Spread through pigs and bats, it causes encephalitis (swelling of the brain) and respiratory infections in humans. For those who catch Nipah encephalitis, CFR could be as high as 61%

RIFT VALLEY FEVER Largely through mosquito bites, it causes flu-like symptoms and high fever in humans. In outbreak areas, CFR is 30%

ZIKA VIRUS: The Aedes mosquito is the vector here. Zika is transmitted from mother to foetus during pregnancy, through sexual contact, blood transfusion and organ transplantation. Non-lethal but high risk to pregnant women as it can cause neurological disorders in the child

and other respiratory disorders in the aftermath of Covid. "Having compromised respiration greatly affects not just your day-to-day activities but, in the long run, also your overall health and wellbeing," he adds.

HOW IS INDIA PREPARING?

India is gearing up for disease X like the rest of the world is. "Even the smallest of labs in India were converted into RT-PCR testing facilities. What Covid taught us about laboratories, beds, medicine, life-saving equipment, vaccines and public awareness must not be forgotten as we will need them in the future," says pulmonologist and former AIIMS director Dr Randeep Guleria. On October 17, Union health secretary Sudhansh Pant pointed out the dire need for the country to have greater knowledge and skill to identify zoonotic diseases. "Seventy five per cent of newly emerging infectious diseases that have affected people in the last 30 years have been zoonotic in nature," he says. India not only has a diverse wildlife but also one of the largest global livestock populations and highest population density, all of which increase the risk of animal-human contact and zoonosis.

Given this new emerging health threat, the government is going the 'One Health' way. Together with the National Centre for Disease Control (NCDC), this initiative aims to improve pandemic preparedness for zoonotic diseases. One of their first steps has been to set up a scientific research collaboration between the National Institute of Animal Biotechnology (NIAB), under the Department of Biotechnology (DBT), and the Bengaluru-based Tata Institute of Genetics and Society on infectious disease diagnostics, surveillance and management in animals.

Research is also indicating the importance of vaccine preparedness. The modelling by Metabiota shows that if one can be rolled out within 100 days of the discovery of the pathogen, the likelihood of a pandemic as deadly as Covid-19 in the next decade drops from 27.5 per cent to 8.1 per cent. In the case of Covid, it took 326 days from the release of the genetic sequence till the first vaccine was authorised. Globally, groups like the Coalition for Epidemic Preparedness Innovations (CEPI), are coming up with initiatives that could see a vaccine rollout within 100 days of a virus with pandemic potential emerging under a $3.5 billion (Rs 29,134 crore) plan.

VACCINE PREPAREDNESS IS CRUCIAL. THE MODELLING SHOWS IF A VACCINE CAN BE ROLLED OUT WITHIN 100 DAYS OF DISCOVERY OF A PATHOGEN, THE LIKELIHOOD OF A PANDEMIC TURNING DEADLY DROPS FROM 27.5% TO 8.1%

In India, the department of pharmaceuticals is working with the US-based Program for Appropriate Technology in Health (PATH) and CEPI to get stakeholders from G20 member-states and a few special invitee countries for a Global Vaccine Research Collaborative. In the recently concluded G20 summit, Union health minister Mansukh Mandaviya stressed the need for such a collaboration at the earliest. "The Indian government has provided financial incentives and streamlined regulatory processes to encourage vaccine manufacturers to increase production capacity. It has also taken steps to ensure the availability of vaccines in rural areas by leveraging the existing infrastructure of primary health centres and other healthcare facilities".

Experts also say that to minimise the risk of a zoonosis outbreak, it is important for us to rethink our relationship with animals. "Humans have never been this close to pathogens from other species," says Dr Mishra. Reforming farming practices, committing to climate change initiatives, reducing deforestation and contact with wild animals will all contribute towards a higher likelihood of Disease X just remaining a paper virus, not a real affliction. ■